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STRETCHABLE DISPLAY DEVICE

Abstract

A stretchable display device includes a display area including a first display area and a second display area adjacent to each other in a first direction; a plurality of rigid portions in the display area and spaced apart from each other in the first direction and a second direction; and a soft portion between adjacent rigid portions, among the plurality of rigid portions, in the first direction and the second direction, wherein the plurality of rigid portions include first, second, and third rigid portions, wherein the first rigid portion is provided in the first display area, and the second rigid portion is provided in the second display area, and wherein the third rigid portion is disposed between the first rigid portion and the second rigid portion and has a larger area than each of the first and second rigid portions.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to Korean Patent Application No. 10-2024-0025748, filed in the Republic of Korea on Feb. 22, 2024, the entire contents of which are hereby expressly incorporated by reference into the present application.

BACKGROUND

Field of the Disclosure

[0002] The present disclosure relates to a display device, and more particularly, to a stretchable display device.

Discussion of the Background Art

[0003] As the information society progresses, interest in displays that process and display a large amount of information has been increasing, and various types of displays have been developed.

[0004] Accordingly, in addition to a commonly known rectangular display, flexible display devices such as a bendable display device for gaming, a foldable display device capable of being folded and unfolded, and a rollable display device having optimal space utilization have been widely developed.

[0005] Recently, a stretchable display device, which is much more flexible than these flexible display devices, has been in the spotlight as a next-generation display.

[0006] The stretchable display device is a display that can freely transform the shape of a screen without distortion even when the size of the screen is increased, folded, or twisted. Unlike the bendable, foldable, or rollable display devices that can only be transformed in a specific area or direction, the stretchable display device is able to implement the ultimate free-form and is considered as the most suitable display for the era of the Internet of Things (IoT), 5G, and autonomous vehicles.

[0007] The stretchable display device may include a rigid portion in which a pixel is disposed and a soft portion in which a connection line connecting the pixels is disposed. The rigid portion may not be stretched, and the soft portion may be stretched.

[0008] Accordingly, in order to secure stretching properties and repeated stretching reliability, a sufficient space for the soft portion is required, which limits the increase in the resolution of the stretchable display device.

SUMMARY

[0009] Accordingly, an object of the present disclosure is to provide a stretchable display device that substantially obviates one or more of the limitations and disadvantages described above and associated with the background art.

[0010] Also, another object of the present disclosure is to provide a stretchable display device with high resolution.

[0011] Additional features and aspects will be set forth in the description that follows, and in part will be apparent from the description, or can be learned by practice of the present disclosure provided herein. Other features and aspects of the inventive concepts can be realized and attained by the structure particularly pointed out in the written description, or derivable therefrom, and the claims hereof as well as the appended drawings.

[0012] To achieve these and other aspects of the present disclosure, as embodied and broadly described herein, a stretchable display device includes a display area including a first display area and a second display area adjacent to each other in a first direction; a plurality of rigid portions in the display area and spaced apart from each other in the first direction and a second direction; and a

soft portion between adjacent rigid portions, among the plurality of rigid portions, in the first direction and the second direction, wherein the plurality of rigid portions include first, second, and third rigid portions, wherein the first rigid portion is provided in the first display area, and the second rigid portion is provided in the second display area, and wherein the third rigid portion is disposed between the first rigid portion and the second rigid portion and has a larger area than each of the first and second rigid portions.
[0013] It is to be understood that both the foregoing general description and the following detailed description are examples and are intended to provide further explanation of the inventive concepts as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0014] The accompanying drawings, which are included to provide a further understanding of the present disclosure and which are incorporated in and constitute a part of this application, illustrate aspects of the disclosure and together with the description serve to explain various principles of the present disclosure. In the drawings:
[0015] FIG. **1** is a schematic plan view of a stretchable display device according to an embodiment of the present disclosure;
[0016] FIG. **2** is a cross-sectional view corresponding to line I-I′ in FIG. **1**;
[0017] FIG. **3** is a schematic plan view enlarging a part of a stretchable display device according to an embodiment of the present disclosure;
[0018] FIG. **4** is an equivalent circuit diagram for a sub-pixel of a stretchable display device according to an embodiment of the present disclosure;
[0019] FIG. **5** is a schematic cross-sectional view along line II-II′ in FIG. **3**;
[0020] FIG. **6** is a schematic plan view of the stretchable display device according to an embodiment of the present disclosure in the folded state; and
[0021] FIG. **7** is a schematic cross-sectional view corresponding to line III-III′ and line IV-IV′ in FIG. **6**.
DETAILED DESCRIPTION
[0022] Advantages and features of the present disclosure and methods for achieving them will be made clear from embodiments described in detail below with reference to the accompanying drawings. The present disclosure can, however, be implemented in many different forms and should not be construed as being limited to the embodiments set forth herein, and the embodiments are provided such that this disclosure will be thorough and complete and will fully convey the scope of the present disclosure to those skilled in the art to which the present disclosure pertains.
[0023] Shapes, sizes, ratios, angles, numbers, and the like disclosed in the drawings for describing embodiments of the present disclosure are merely illustrative examples, and thus the present disclosure is not limited to the illustrated examples. The same reference numerals refer to the same components throughout this disclosure unless otherwise specified. Further, in the following description of the present disclosure, where a detailed description of a known related art may unnecessarily obscure the gist of the present disclosure, the detailed description thereof may be omitted herein or may be briefly discussed.
[0024] Where terms such as “including,” “having,” “comprising,” and the like are used in this disclosure, other parts can be added unless a more limiting term like “only” is used herein. Further, where a component is expressed as being singular, being plural is included, and vice versa, unless otherwise specified.
[0025] In analyzing a component, an error range should be interpreted as being included even where there is no explicit description.

[0026] In describing a positional relationship, for example, where a positional relationship of two parts/layers is described as being "over," "on," "above," "below," "under," "next to," or the like, one or more other parts/layers can be provided between the two parts/layers, unless a more limiting term like "immediately" or "directly" is used therewith.
[0027] In describing a temporal relationship, for example, where a temporal predecessor relationship is described as being "after," "subsequent," "next to," "prior to," or the like, unless a more limiting term like "immediately" or "directly" is used, cases that are not continuous or sequential can also be included.
[0028] Although the terms first, second, and the like may be used to describe various components, these components are not substantially limited by these terms. These terms are used only to refer to one component separately from another component, and may not define any particular order or sequence. Therefore, a first component described below can substantially be a second component, and vice versa, within the technical spirit of the present disclosure.
[0029] Features of various embodiments of the present disclosure can be partially or entirely united or combined with each other, technically various interlocking and driving are possible, and each of the embodiments can be independently implemented with respect to each other or implemented together in a co-dependent relationship.
[0030] Hereinafter, example embodiments of the present disclosure will be described in detail with reference to accompanying drawings.
[0031] FIG. **1** is a schematic plan view of a stretchable display device according to an embodiment of the present disclosure.
[0032] In FIG. **1**, a stretchable display device according to an embodiment of the present disclosure may include a rigid portion A**1** and a soft portion A**2** provided in a display area DA and may be stretched in a first direction X and/or a second direction. Here, the rigid portion A**1** may not be stretched, and the soft portion A**2** may be stretched.
[0033] The rigid portion A**1** may be provided in the form of an island, and a plurality of rigid portions A**1** may be disposed to be spaced apart from each other along the first direction X and the second direction Y. For example, the rigid portion A**1** may have a polygonal shape. The rigid portions A**1** may be arranged in a matrix form.
[0034] A pixel including a plurality of sub-pixels may be provided in the rigid portion A**1**. Each of the plurality of sub-pixels may include a light-emitting element, at least one thin film transistor, a plurality of signal lines, and a plurality of signal electrodes.
[0035] The soft portion A**2** may be disposed between adjacent rigid portions A**1** in each of the first direction X and the second direction Y. The soft portion A**2** may include a horizontal soft portion A**21** and a vertical soft portion A**22**. The horizontal soft portion A**21** may extend substantially in the first direction X, and the vertical soft portion A**22** may extend substantially in the second direction Y. Here, the horizontal soft portion A**21** may be disposed between the adjacent rigid portions A**1** in the first direction X, and the vertical soft portion A**22** may be disposed between the adjacent rigid portions A**1** in the second direction Y.
[0036] A stretchable line that is a connection line connecting the adjacent pixels may be provided in the soft portion A**2**. The stretchable line may include a plurality of voltage lines such as a gate line, a data line, a high potential line, a low potential line, an emission line, and a reference voltage line. The stretchable line may have at least one curved part.
[0037] The display area DA of the stretchable display device according to the embodiment of the present disclosure may include a first display area DA**1** and a second display area DA**2**. The first display area DA**1** and the second display area DA**2** may be adjacent to each other in the first direction X.
[0038] A first rigid portion A**11** may be provided in the first display area DA**1**, and a second rigid portion A**12** may be provided in the second display area DA**2**.
[0039] Here, the first rigid portion A**11** may be disposed on the same line as the second rigid

portion A**12** in the first direction X. That is, the first rigid portion A**11** and the second rigid portion A**12** may be disposed on different lines in the first direction X. Specifically, an imaginary straight line L**1** passing through the centers of the first rigid portions A**11** in the first direction X may not coincide with an imaginary straight line L**2** passing through the centers of the second rigid portions A**12** in the first direction X, and the imaginary straight lines L**1** and L**2** may be spaced apart from each other in the second direction Y.

[0040] In addition, a third rigid portion A**13** may be further provided between the first rigid portion A**11** and the second rigid portion A**12**. The third rigid portion A**13** may be disposed in the second display area DA**2**.

[0041] The first rigid portion A**11** and the second rigid portion A**12** may have substantially the same shape, and the third rigid portion A**13** may have a different shape from the first and second rigid portions A**11** and A**12**. For example, the first and second rigid portions A**11** and A**12** may have a substantially square shape, and the third rigid portion A**13** may have an L-like shape. However, embodiments of the present disclosure are not limited thereto, and the first, second, and third rigid portions A**11**, A**12**, and A**13** may have various shapes.

[0042] Further, the area of the third rigid portion A**13** may be greater than the area of each of the first and second rigid portions A**11** and A**12**. In this case, the first and second rigid portions A**11** and A**12** may have the same shape.

[0043] A cross-sectional structure of a stretchable display device according to the embodiment of the present disclosure will be described in detail with reference to FIG. **2**.

[0044] FIG. **2** is a cross-sectional view corresponding to line I-I′ in FIG. **1**.

[0045] In FIG. **2**, the stretchable display device **100** according to the embodiment of the present disclosure may include a first flexible substrate FS**1** and a second flexible substrate FS**2**. The rigid portions A**1** and the soft portions A**2** may be provided between the first flexible substrate FS**1** and the second flexible substrate FS**2**. The first flexible substrate FS**1** and the second flexible substrate FS**2** may be attached to the rigid portions A**1** and the soft portions A**2** through a first adhesive layer and a second adhesive layer, respectively.

[0046] The first flexible substrate FS**1** and the second flexible substrate FS**2** may be formed of a soft matter or soft material with bending or stretching properties. For example, the first flexible substrate FS**1** and the second flexible substrate FS**2** may be formed of silicone rubber such as polydimethylsiloxane (PDMS), elastomer such as polyurethane (PU), or styrene butadiene block copolymer such as styrene butadiene styrene (SBS).

[0047] The first flexible substrate FS**1** and the second flexible substrate FS**2** may be formed of the same material. However, embodiments of the present disclosure are not limited thereto. In other embodiments, the first flexible substrate FS**1** and the second flexible substrate FS**2** may be formed of different materials.

[0048] The first flexible substrate FS**1** and the second flexible substrate FS**2** may have relatively low elastic modulus, that is, Young's modulus, and may have a relatively high ductile breaking rate. Here, the elastic modulus is a value representing the rate of deformation relative to the stress applied to an object. If the elastic modulus is relatively high, the hardness may be relatively high. In addition, the ductile breaking rate refers to the elongation rate at the point when the stretched object is broken or cracked.

[0049] For example, each of the first flexible substrate FS**1** and the second flexible substrate FS**2** may have the elastic modulus of several MPa to hundreds of MPa and the ductile breaking rate of about 100% or more. In addition, each of the first flexible substrate FS**1** and the second flexible substrate FS**2** may have a thickness of about 10 μm to about 1 mm. However, embodiments of the present disclosure are not limited thereto.

[0050] In addition, the first and second adhesive layers may be formed of an acryl-based, silicon-based, or urethane-based adhesive. For example, the first and second adhesive layers may be optically clear adhesive (OCA) that is formed and attached in the form of a film or optically clear

resin (OCR) that is cured after applying a liquid material.
[0051] As described above, the display area DA may include the first display area DA**1** and the second display area DA**2**. The plurality of first rigid portions A**11** may be provided in the first display area DA**1**, the plurality of second rigid portions A**12** may be provided in the second display area DA**2**, and the third rigid portion A**13** may be provided between the first rigid portion A**11** and the second rigid portion A**12**. The third rigid portion A**13** may be disposed in the second display area DA**2**. However, embodiments of the present disclosure are not limited thereto. In other embodiments, the third rigid portion A**13** may be disposed in the first display area DA**1**.
[0052] The soft portion A**2** may be provided between the adjacent first, second, and third rigid portions A**11**, A**12**, and A**13**. Specifically, the soft portion A**2** may be provided between the adjacent first rigid portions A**11**, between the adjacent second rigid portions A**12**, between the adjacent first and third rigid portions A**11** and A**13**, and between the adjacent second and third rigid portions A**12** and A**13**.
[0053] The configurations of the first, second, and third rigid portions A**11**, A**12**, and A**13** will be described in detail with reference to FIG. **3**.
[0054] FIG. **3** is a schematic plan view enlarging a part of a stretchable display device according to the embodiment of the present disclosure.
[0055] In FIG. **3**, the stretchable display device according to the embodiment of the present disclosure may include the rigid portion A**1** and the soft portion A**2**. The rigid portion A**1** may include the first, second, and third rigid portions A**11**, A**12**, and A**13**, and the third rigid portion A**13** may be disposed between the first and second rigid portions A**11** and A**12**.
[0056] The pixel including the plurality of sub-pixels SP**1**, SP**2**, and SP**3** may be provided in each of the first, second, and third rigid portions A**11**, A**12**, and A**13**. For example, first, second, and third sub-pixels SP**1**, SP**2**, and SP**3** may be provided in each of the first, second, and third rigid portions A**11**, A**12**, and A**13**, and the first, second, and third sub-pixels SP**1**, SP**2**, and SP**3** may be red, green, and blue sub-pixels, respectively. However, embodiments of the present disclosure are not limited thereto. In other embodiments, the first, second, and third sub-pixels SP**1**, SP**2**, and SP**3** may be blue, green, and red sub-pixels, respectively.
[0057] Each of the first, second, and third sub-pixels SP**1**, SP**2**, and SP**3** may include a light-emitting element, at least one thin film transistor, and at least one capacitor.
[0058] The soft portion A**2** may be provided between the adjacent rigid portions A**1** in each of the first direction X and the second direction Y, that is, between the adjacent first, second, and third rigid portions A**11**, A**12**, and A**13** in each of the first direction X and the second direction Y. Specifically, the soft portion A**2** may be provided between the adjacent first rigid portions A**11** in each of the first direction X and the second direction Y, between the adjacent second rigid portions A**12** in each of the first direction X and the second direction Y, between the adjacent third rigid portions A**13** in the second direction Y, between the adjacent first and third rigid portions A**11** and A**13** in the first direction X, and between the adjacent second and third rigid portions A**12** and A**13** in the first direction X.
[0059] The soft portion A**2** may include the horizontal soft portion A**21** extending substantially in the first direction X and the vertical soft portion A**22** extending substantially in the second direction Y.
[0060] The stretchable line may be provided in the soft portion A**2**. The stretchable line may have at least one curved part. For example, the stretchable line may have a wave structure and may include a plurality of wave shapes.
[0061] In the first direction X or the second direction Y, one rigid portion A**1** and one soft portion A**2** may form a unit, and a sum of a first length a**1** of the rigid portion A**1** and a second length a**2** of the soft portion A**2** may become a pitch p**1** of one unit. Here, the first length a**1** and the second length a**2** may be equal to each other. That is, the first length a**1** and the second length a**2** may be ½ pitch, i.e., (p**1**)/2. However, embodiments of the present disclosure are not limited thereto. In other

embodiments, the first length a**1** may be smaller than the second length a**2**.
[0062] As described above, the first rigid portion A**11** and the second rigid portion A**12** in the first direction X may be disposed on different lines. In this case, the first rigid portion A**11** and the second rigid portion A**12** may be spaced apart by ½ pitch (p**1**)/2 in the second direction Y.
[0063] Meanwhile, the third rigid portion A**13** may include an emission portion A**1***a* and a connection portion A**1***b*. The first, second, and third sub-pixels SP**1**, SP**2**, and SP**3** may be provided in the emission portion A**1***a*, and a plurality of signal lines may be provided in the connection portion A**1***b*.
[0064] Here, the emission portion A**1***a* of the third rigid portion A**13** may have the same lengths of the first direction X and the second direction Y. On the other hand, the connection portion A**1***b* of the third rigid portion A**13** may have a length of the first direction X and a length of the second direction Y greater than the length of the first direction X, and the length of the second direction Y may be substantially equal to or smaller than the pitch p**1**.
[0065] The emission portion A**1***a* of the third rigid portion A**13** may have substantially the same shape as the first and second rigid portions A**11** and A**12**. In addition, the emission portion A**1***a* may have substantially the same configuration as the first and second rigid portions A**11** and A**12**.
[0066] The emission portion A**1***a* of the third rigid portion A**13** may be disposed on the same line as the second rigid portion A**12** in the first direction X and may be disposed on a different line from the first rigid portion A**11**.
[0067] On the other hand, the connection portion A**1***b* of the third rigid portion A**13** may have a length of the first direction X smaller than the first length a**1** of the first and second rigid portions A**11** and A**12** and a length of the second direction Y greater than the first length a**1** of the first and second rigid portions A**11** and A**12**. In addition, the length of the second direction Y may be equal to or smaller than the pitch p**1** and may be greater than ½ pitch (p**1**)/2. Accordingly, in the first direction X, the length of the connection portion A**1***b* may be smaller than the length of the emission portion A**1***a*, and in the second direction Y, the length of the connection portion A**1***b* may be greater than the length of the emission portion A**1***a*.
[0068] The third rigid portion A**13** may have a substantially L-like shape. The connection portions A**1***b* of the adjacent third rigid portions A**13** in the second direction Y may be spaced apart from each other or may be in contact with each other. Preferably, the length of the connection portion A**1***b* of the third rigid portion A**13** in the second direction Y may be similar to the pitch p**1**. When the display device is not stretched in the second direction Y, the connection portions A**1***b* of the adjacent third rigid portions A**13** in the second direction Y may be in contact with each other, and when the display device is stretched in the second direction Y, the connection portions A**1***b* of the adjacent third rigid portions A**13** in the second direction Y may be spaced apart from each other. The area of the third rigid portion A**13** may be greater than the area of the first and second rigid portions A**11** and A**12**.
[0069] The configuration of the sub-pixel provided in the first, second, and third rigid portions A**11**, A**12**, and A**13** may be described with reference to FIG. **4**.
[0070] FIG. **4** is an equivalent circuit diagram for a sub-pixel of a stretchable display device according to an embodiment of the present disclosure.
[0071] In FIG. **4**, one sub-pixel of the stretchable display device according to the embodiment of the present disclosure, that is, each of the first, second, and third sub-pixels SP**1**, SP**2**, and SP**3** of FIG. **3** may include a driving transistor DT, first, second, third, fourth, and fifth transistors T**1**, T**2**, T**3**, T**4**, and T**5**, a storage capacitor Cst, and a light-emitting diode LED.
[0072] For example, the driving transistor DT and the first, second, third, fourth, and fifth transistors T**1**, T**2**, T**3**, T**4**, and T**5** may be P-type transistors. However, embodiments of the present disclosure are not limited thereto. In other embodiments, the driving transistor DT and the first, second, third, fourth, and fifth transistors T**1**, T**2**, T**3**, T**4**, and T**5** may be N-type transistors.
[0073] The driving transistor DT may be switched according to a voltage of a first capacitor

electrode of the storage capacitor Cst and may be connected to a high potential voltage ELVDD. Specifically, a gate of the driving transistor DT may be connected to the first capacitor electrode of the storage capacitor Cst and a source of the second transistor T**2**. A source of the driving transistor DT may be connected to the high potential voltage ELVDD. A drain of the driving transistor DT may be connected to a drain of the second transistor T**2** and a source of the fourth transistor T**4**.
[0074] The first transistor T**1** may be switched according to a gate signal SCAN and may be connected to a data signal Vdata. Specifically, a gate of the first transistor T**1** may be connected to the gate signal SCAN. A source of the first transistor T**1** may be connected to the data signal Vdata. A drain of the first transistor T**1** may be connected to a second capacitor electrode of the storage capacitor Cst and a source of the third transistor T**3**.
[0075] The second transistor T**2** may be switched according to the gate signal SCAN and may be connected to the driving transistor DT. Specifically, a gate of the second transistor T**2** may be connected to the scan signal SCAN. The source of the second transistor T**2** may be connected to the first capacitor electrode of the storage capacitor Cst and the gate of the driving transistor DT. The drain of the second transistor T**2** may be connected to the source of the driving transistor DT and the source of the fourth transistor T**4**.
[0076] The third transistor T**3** may be switched according to an emission signal EM and may be connected to a reference voltage Vref. A gate of the third transistor T**3** may be connected to the emission signal EM. The source of the third transistor T**3** may be connected to the second capacitor electrode of the storage capacitor Cst and the drain of the first transistor T**1**. A drain of the third transistor T**3** may be connected to the reference voltage Vref and a source of the fifth transistor T**5**.
[0077] The fourth transistor T**4** may be switched according to the emission signal EM and may be connected to the driving transistor DT and the light-emitting diode LED. Specifically, a gate of the fourth transistor T**4** may be connected to the emission signal EM. The source of the fourth transistor T**4** may be connected to the drain of the driving transistor DT and the drain of the second transistor T**2**. A drain of the fourth transistor T**4** may be connected to a drain of the fifth transistor T**5** and a first electrode of the light-emitting diode LED.
[0078] The fifth transistor T**5** may be switched according to the gate signal SCAN and may be connected to the reference voltage Vref and the fourth transistor T**4**. Specifically, a gate of the fifth transistor T**5** may be connected to the scan signal SCAN. The source of the fifth transistor T**5** may be connected to the reference voltage Vref and the drain of the third transistor T**3**. The drain of the fifth transistor T**5** may be connected to the drain of the fourth transistor T**4** and the first electrode of the light-emitting diode LED.
[0079] The storage capacitor Cst may store the data signal Vdata and a threshold voltage Vth of the driving transistor DT. The first capacitor electrode of the storage capacitor Cst may be connected to the gate of the driving transistor DT and the source of the second transistor T**2**. The second capacitor electrode of the storage capacitor Cst may be connected to the drain of the first transistor T**1** and the source of the third transistor T**3**.
[0080] The light-emitting diode LED may be connected between the fourth and fifth transistors T**4** and T**5** and a low potential voltage ELVSS and may emit light with luminance proportional to a current of the driving transistor DT. The first electrode of the light-emitting diode LED, which is an anode, may be connected to the drain of the fourth transistor T**4** and the drain of the fifth transistor T**5**. The second electrode of the light-emitting diode LED, which is a cathode, may be connected to the low potential voltage ELVSS.
[0081] In the embodiment of the present disclosure of FIG. **4**, as an example, each sub-pixel has a 6T1C structure including six transistors and one capacitor, but in other embodiments, each sub-pixel may have one of 2T1C, 4T1C, 5T1C, 3T2C, 4T2C, 5T2C, 6T2C, 7T1C, 7T2C, 8T1C, and 8T2C structures.
[0082] A cross-sectional structure of a stretchable display device according to an embodiment of the present disclosure will be described in detail with reference to FIG. **5**.

[0083] FIG. **5** is a schematic cross-sectional view of a stretchable display device according to an embodiment of the present disclosure. FIG. **5** shows a cross-section corresponding to line II-II′ in FIG. **3**.
[0084] In FIG. **5**, the stretchable display device according to the embodiment of the present disclosure may include a base substrate **110** provided with the rigid portion A**1** and the soft portion A**2**.
[0085] The base substrate **110** may include a first base portion **10***a* and a second base portion **110***b*. The first base portion **110***a* may be disposed to correspond to the rigid portion A**1**, and the second base portion **110***b* may be disposed to correspond to the soft portion A**2**.
[0086] The first base portion **110***a* may be provided in a plate shape in a display area and may serve to support and protect components of the plurality of sub-pixels SP**1**, SP**2**, and SP**3**. The first base portion **110***a* may be plural, and the plurality of first base portions **110***a* may be spaced apart from each other in the first direction X and the second direction Y.
[0087] The second base portion **110***b* may be provided between adjacent first base portions **110***a* in the first direction X and the second direction Y. The second base portion **110***b* may include at least one curved part and may serve to support and protect the stretchable line **134**. The second base portion **110***b* may have substantially the same shape as the stretchable line **134**. The first and second base portions **110***a* and **110***b* may be connected to each other and may be provided as one-body.
[0088] Meanwhile, the first flexible substrate FS**1** of FIG. **2** may be provided under the base substrate **110**. The first flexible substrate FS**1** may be attached to the bottom surface of the base substrate **110** through the first adhesive layer.
[0089] The base substrate **110** may be formed of a rigid material having lower flexibility than the soft material of the first flexible substrate FS**1**. For example, the base substrate **110** may be formed of a polyimide (PI) resin or epoxy resin.
[0090] The base substrate **110** may have relatively high elastic modulus, and the elastic modulus of the base substrate **110** may be higher than the elastic modulus of the first flexible substrate FS**1**. For example, the elastic modulus of the base substrate **110** may be more than 1,000 times higher than the elastic modulus of the first flexible substrate FS**1**, but embodiments of the present disclosure are not limited thereto.
[0091] A first buffer layer **111** may be provided on the base substrate **110** as a first insulation layer. The first buffer layer **111** may block permeation of moisture or oxygen from the outside to protect the components of the plurality of sub-pixels SP**1**, SP**2**, and SP**3**.
[0092] The first buffer layer **111** may be formed as a single layer or multiple layers of an inorganic insulating material. The inorganic insulating material of the first buffer layer **111** may include silicon nitride (SiNx), silicon oxide (SiOx), or silicon oxynitride (SiON).
[0093] To prevent or reduce damage of the first buffer layer **111** such as cracks due to stretching, the first buffer layer **111** may be removed in the soft portion A**2** to substantially correspond to the rigid portion A**1**, so that the first buffer layer **111** may be provided over the first base portion **110***a* of the base substrate **110** and may not be provided over the second base portion **110***b*.
[0094] In other embodiments, the first buffer layer **111** may be omitted.
[0095] A light shielding layer **121** may be provided on the first buffer layer **111** of the rigid portion A**1**. The light shielding layer **121** may be formed of a conductive material such as metal. For example, the light shielding layer **121** may be formed of at least one of aluminum (Al), copper (Cu), molybdenum (Mo), titanium (Ti), chromium (Cr), nickel (Ni), tungsten (W), or an alloy thereof. The light shielding layer **121** may have a single-layered structure or a multiple-layered structure.
[0096] A second buffer layer **112** may be provided on the light shielding layer **121** as a second insulation layer. The second buffer layer **112** may be formed as a single layer or multiple layers of an inorganic insulating material. The inorganic insulating material of the second buffer layer **112**

may include silicon nitride (SiNx), silicon oxide (SiOx), or silicon oxynitride (SiON).
[0097] To prevent or reduce damage of the second buffer layer **112** such as cracks due to stretching, the second buffer layer **112** may be removed in the soft portion A**2** to substantially correspond to the rigid portion A**1**. The second buffer layer **112** may be provided over the first base portion **110***a* of the base substrate **110** and may not be provided over the second base portion **110***b*.
[0098] A semiconductor layer **122** may be provided on the second buffer layer **112**. The semiconductor layer **122** may overlap the light shielding layer **121**, and the light shielding layer **121** may block light incident on the semiconductor layer **122** and prevent the semiconductor layer **122** from deteriorating due to the light.
[0099] The semiconductor layer **122** may include a channel region at its central part and source and drain regions at both sides of the channel region.
[0100] The semiconductor layer **122** may be formed of an oxide semiconductor material. Alternatively, the semiconductor layer **122** may be formed of polycrystalline silicon, and in this case, both ends of the semiconductor layer **122** may be doped with impurities.
[0101] A gate insulation layer **113** may be provided on the semiconductor layer **122** as a third insulation layer. The gate insulation layer **113** may be formed as a single layer or multiple layers of an inorganic insulating material. The inorganic insulating material of the gate insulation layer **113** may include silicon nitride (SiNx), silicon oxide (SiOx), or silicon oxynitride (SiON).
[0102] To prevent or reduce damage of the gate insulation layer **113** such as cracks due to stretching, the gate insulation layer **113** may be removed in the soft portion A**2** to substantially correspond to the rigid portion A**1**. The gate insulation layer **113** may be provided over the first base portion **110***a* of the base substrate **110** and may not be provided over the second base portion **110***b*.
[0103] A gate electrode **123** and a first connection electrode **124** may be provided on the gate insulation layer **113**.
[0104] The gate electrode **123** may overlap the semiconductor layer **122** and may be disposed to correspond to the central part of the semiconductor layer **122**. Accordingly, the gate electrode **123** may also overlap the light shielding layer **121**.
[0105] The first connection electrode **124** may be spaced apart from the semiconductor layer **122** and may overlap the light shielding layer **121**. The first connection electrode **124** may be in contact with the light shielding layer **121** through a contact hole provided in the second buffer layer **112** and the gate insulation layer **113**.
[0106] The gate electrode **123** and the first connection electrode **124** may be formed of a conductive material such as metal. For example, the gate electrode **123** and the first connection electrode **124** may be formed of at least one of aluminum (Al), copper (Cu), molybdenum (Mo), titanium (Ti), chromium (Cr), nickel (Ni), tungsten (W), or an alloy thereof. The gate electrode **123** and the first connection electrode **124** may have a single-layered structure or a multiple-layered structure.
[0107] A first interlayer insulation layer **114** may be provided on the gate electrode **123** and the first connection electrode **124** as a fourth insulation layer. The first interlayer insulation layer **114** may be formed as a single layer or multiple layers of an inorganic insulating material. The inorganic insulating material of the first interlayer insulation layer **114** may include silicon nitride (SiNx), silicon oxide (SiOx), or silicon oxynitride (SiON).
[0108] To prevent or reduce damage of the first interlayer insulation layer **114** such as cracks due to stretching, the first interlayer insulation layer **114** may be removed in the soft portion A**2** to substantially correspond to the rigid portion A**1**. The first interlayer insulation layer **114** may be provided over the first base portion **110***a* of the base substrate **110** and may not be provided over the second base portion **110***b*.
[0109] An auxiliary electrode **125**, an auxiliary line **126**, and a pad electrode **127** may be provided on the first interlayer insulation layer **114**. The auxiliary electrode **125** may overlap gate electrode

123, the semiconductor layer **122**, and the light shielding layer **121**. The auxiliary line **126** may overlap the light shielding layer **121** and may be spaced apart from the gate electrode **123**, the semiconductor layer **122**, and the first connection electrode **124**. The pad electrode **127** may be spaced apart from the light shielding layer **121** and may be disposed close to an edge of the rigid portion A**1** adjacent thereto.

[0110] The auxiliary electrode **125**, the auxiliary line **126**, and the pad electrode **127** may be formed of a conductive material such as metal. For example, the auxiliary electrode **125**, the auxiliary line **126**, and the pad electrode **127** may be formed of at least one of aluminum (Al), copper (Cu), molybdenum (Mo), titanium (Ti), chromium (Cr), nickel (Ni), tungsten (W), or an alloy thereof. The auxiliary electrode **125**, the auxiliary line **126**, and the pad electrode **127** may have a single-layered structure or a multiple-layered structure.

[0111] A second interlayer insulation layer **115** may be provided on the auxiliary electrode **125**, the auxiliary line **126**, and the pad electrode **127** as a fifth insulation layer. The second interlayer insulation layer **115** may be formed as a single layer or multiple layers of an inorganic insulating material. The inorganic insulating material of the second interlayer insulation layer **115** may include silicon nitride (SiNx), silicon oxide (SiOx), or silicon oxynitride (SiON).

[0112] To prevent or reduce damage of the second interlayer insulation layer **115** such as cracks due to stretching, the second interlayer insulation layer **115** may be removed in the soft portion A**2** to substantially correspond to the rigid portion A**1**. The second interlayer insulation layer **115** may be provided over the first base portion **110***a* of the base substrate **110** and may not be provided over the second base portion **110***b*.

[0113] A source electrode **128** a drain electrode **129**, a second connection electrode **131**, and a power line **132** may be provided on the second interlayer insulation layer **115**.

[0114] The source electrode **128** and the drain electrode **129** may be spaced apart from each other with the gate electrode **123** positioned therebetween and may be in contact with both ends of the semiconductor layer **122** through contact holes provided in the first and second interlayer insulation layers **114** and **115** and the gate insulation layer **113**. The gate electrode **123** and the auxiliary electrode **125** may be disposed between the source electrode **128** and the drain electrode **129**.

[0115] The semiconductor layer **122**, the gate electrode **123**, the source electrode **128**, and the drain electrode **129** may constitute a thin film transistor TR.

[0116] The second connection electrode **131** may be spaced apart from the thin film transistor TR. The second connection electrode **131** may overlap the first connection electrode **124** and may be in contact with the first connection electrode **124** through a contact hole provided in the first and second interlayer insulation layers **114** and **115**. In addition, the second connection electrode **131** may overlap the light shielding layer **121**.

[0117] The power line **132** may be spaced apart from the thin film transistor TR. The power line **132** may overlap the auxiliary line **126** and may be in contact with the auxiliary line **126** through a contact hole formed in the second interlayer insulation layer **115**.

[0118] For example, the power line **132** may be a line supplying the low potential voltage ELVSS. In this case, the power line **132** or the auxiliary line **126** may be connected to the light shielding layer **121**. That is, the light shielding layer **121** may be supplied with the low potential voltage ELVSS.

[0119] The source electrode **128**, the drain electrode **129**, the second connection electrode **131**, and the power line **132** may be formed of a conductive material such as metal. For example, the source electrode **128**, the drain electrode **129**, the second connection electrode **131**, and the power line **132** may be formed of at least one of aluminum (Al), copper (Cu), molybdenum (Mo), titanium (Ti), chromium (Cr), nickel (Ni), tungsten (W), or an alloy thereof. The source electrode **128**, the drain electrode **129**, the second connection electrode **131**, and the power line **132** may have a single-layered structure or a multiple-layered structure.

[0120] Next, a third interlayer insulation layer **116** may be provided on the source electrode **128**,

the drain electrode **129**, the second connection electrode **131**, and the power line **132** as a sixth insulation layer. The third interlayer insulation layer **116** may be formed as a single layer or multiple layers of an inorganic insulating material. The inorganic insulating material of the third interlayer insulation layer **116** may include silicon nitride (SiNx), silicon oxide (SiOx), or silicon oxynitride (SiON).
[0121] To prevent or reduce damage of the third interlayer insulation layer **116** such as cracks due to stretching, the third interlayer insulation layer **116** may be removed in the soft portion A**2** to substantially correspond to the rigid portion A**1**. The third interlayer insulation layer **116** may be provided over the first base portion **110***a* of the base substrate **110** and may not be provided over the second base portion **110***b*.
[0122] Next, an auxiliary pad **133** may be provided on the third interlayer insulation layer **116**. The auxiliary pad **133** may overlap the pad electrode **127** and may be in contact with the pad electrode **127** through a contact hole provided in the second and third interlayer insulation layers **115** and **116**.
[0123] The auxiliary pad **133** may be formed of a conductive material such as metal. For example, the auxiliary pad **133** may be formed of at least one of aluminum (Al), copper (Cu), molybdenum (Mo), titanium (Ti), chromium (Cr), nickel (Ni), tungsten (W), or an alloy thereof. The auxiliary pad **133** may have a single-layered structure or a multiple-layered structure.
[0124] A passivation layer **117** may be provided on the auxiliary pad **133**. The passivation layer **117** may be formed as a single layer or multiple layers of an inorganic insulating material. The inorganic insulating material of the passivation layer **117** may include silicon nitride (SiNx), silicon oxide (SiOx), or silicon oxynitride (SiON).
[0125] To prevent or reduce damage of the passivation layer **117** such as cracks due to stretching, the passivation layer **117** may be removed in the soft portion A**2** to substantially correspond to the rigid portion A**1**. The passivation layer **117** may be provided over the first base portion **110***a* of the base substrate **110** and may not be provided over the second base portion **110***b*.
[0126] In this case, edges of the passivation layer **117** and the third interlayer insulation layer **116** of the rigid portion A**1** adjacent to the soft portion A**2** may be partially removed, thereby exposing a top surface of the second interlayer insulation layer **115**.
[0127] The passivation layer **117** may be omitted.
[0128] A planarization layer **118** may be provided on the passivation layer **117**. The planarization layer **118** may eliminate a step difference due to the layers thereunder and may have a substantially flat top surface. The planarization layer **118** may be formed of an organic insulating material such as photosensitive acrylic polymer (photo acryl).
[0129] The planarization layer **118** may be provided in the rigid portion A**1** and may not be provided in the soft portion A**2**. Accordingly, the planarization layer **118** may be provided over the first base portion **110***a* of the base substrate **110** and may not be provided over the second base portion **110***b*.
[0130] In the rigid portion A**1**, the planarization layer **118** may be in contact with side surfaces of the third interlayer insulation layer **116** and the passivation layer **117** and be in contact with the exposed top surface of the second interlayer insulation layer **115**.
[0131] A first electrode **135**, a second electrode **136**, and the stretchable line **134** may be provided on the planarization layer **118**. The first electrode **135**, the second electrode **136**, and the stretchable line **134** may be formed of a conductive material such as metal. For example, the first electrode **135**, the second electrode **136**, and the stretchable line **134** may be formed of at least one of aluminum (Al), copper (Cu), molybdenum (Mo), titanium (Ti), chromium (Cr), nickel (Ni), tungsten (W), or an alloy thereof. The first electrode **135**, the second electrode **136**, and the stretchable line **134** may have a single-layered structure or a multiple-layered structure.
[0132] The first electrode **135** may overlap the drain electrode **129** and may be in contact with the drain electrode **129** through a contact hole provided in the planarization layer **118**, the passivation

layer **117**, and the third interlayer insulation layer **116**. The second electrode **136** may overlap the second connection electrode **131** and may be in contact with the second connection electrode **131** through a contact hole provided in the planarization layer **118**, the passivation layer **117**, and the third interlayer insulation layer **116**.

[0133] An end of the stretchable line **134** may be disposed on the planarization layer **118** of the rigid portion A**1**. The end of the stretchable line **134** may overlap the auxiliary pad **133** and may be in contact with the auxiliary pad **133** through a contact hole provided in the planarization layer **118** and the passivation layer **117**. The end of the stretchable line **134** may also overlap the pad electrode **127**.

[0134] The stretchable line **134** may extend into and be provided in the soft portion A**2**. The stretchable line **134** may be in contact with top and side surfaces of the planarization layer **160** in the rigid portion A**1** and may be in contact with a top surface of the second base portion **110***b* in the soft portion A**2**. The stretchable line **134** may be in contact with side surfaces of the first buffer layer **111**, the second buffer layer **112**, the gate insulation layer **113**, the first interlayer insulation layer **114**, and the second interlayer insulation layer **115**.

[0135] Meanwhile, a bank layer may be further provided on the first electrode **135**, the second electrode **136**, and the stretchable line **134**. The bank layer may expose at least parts of the first electrode **135** and the second electrode **136** and may cover the end of the stretchable line **134**.

[0136] Next, an adhesive layer **140** may be provided on the first and second electrodes **135** and **136** of the rigid portion A**1**. The adhesive layer **140** may be an anisotropic conductive film (ACF) including an insulating base member and a plurality of conductive balls **142** dispersed in the insulating base member, among others.

[0137] When heat or pressure is applied to the adhesive layer **140**, the conductive balls **142** may be electrically connected in an area where the heat or pressure is applied, so that the adhesive layer **140** may have a conductive property. In an area where the heat or pressure is not applied, the adhesive layer **140** may have an insulating property.

[0138] The light-emitting element **150** may be provided on the adhesive layer **140**. The light-emitting element **150** may be provided in the form of a micro light-emitting diode chip (micro LED chip or uLED chip) including a p-type layer **151**, an active layer **152**, an n-type layer **153**, a p-electrode **157**, and an n-electrode **158**. In addition, the light-emitting element **150** may further include a first ohmic layer **154**, a second ohmic layer **155**, and a protection layer **156**.

[0139] Specifically, the first ohmic layer **154** and the p-electrode **157** may be sequentially provided on one side of a bottom surface of the p-type layer **151**, and the active layer **152**, the n-type layer **153**, the second ohmic layer **155**, and the n-electrode **158** may be spaced apart from the first ohmic layer **154** and sequentially provided on another side of the bottom surface of the p-type layer **151**.

[0140] Here, the protection layer **156** may be provided between the first ohmic layer **154** and the p-electrode **157** and between the second ohmic layer **155** and the n-electrode **158**. The p-electrode **157** may be in contact with the first ohmic layer **154** through a contact hole formed in the protection layer **156**, and the n-electrode **158** may be in contact with the second ohmic layer **155** through a contact hole formed in the protection layer **156**. The protection layer **156** may cover and be in contact with a side surface of the first ohmic layer **154** and may cover and be in contact with side surfaces of the active layer **152**, the n-type layer **153**, and the second ohmic layer **155**. In addition, the protection layer **156** may be in contact with the bottom surface of the p-type layer **151** between the first ohmic layer **154** and the active layer **152**.

[0141] The p-electrode **157** of the light-emitting element **150** may be in contact with the first electrode **135** through the conductive balls **142**, and the n-electrode **158** may be in contact with the second electrode **136** through the conductive balls **142**. Accordingly, the p-electrode **157** of the light-emitting layer **150** may be connected to the drain electrode **129** of the thin film transistor TR through the first electrode **135**, and the n-electrode **158** may be connected to the low potential voltage ELVSS through the second electrode **136**.

[0142] The light-emitting element **150** may have a flip-chip structure in which the p-electrode **157** and the n-electrode **158** are provided on the same side (for example, a side facing the base substrate **110**) and light is emitted through a side opposite to the side provided with the p-electrode **157** and the n-electrode **158** (for example, a top surface of the p-type layer **151**). However, embodiments of the present disclosure are not limited thereto.
[0143] Alternatively, the light-emitting element **150** may have a lateral structure in which the n-electrode and the p-electrode are provided on the same side and light is emitted through the same side provided with the n-electrode and the p-electrode or a vertical structure in which the n-electrode and the p-electrode are provided on opposite sides.
[0144] In addition, the light-emitting element **150** may emit light through the side facing the base substrate **110** as well as the side opposite to the side facing the base substrate **110**. That is, the light-emitting element **150** may be a double-sided light-emitting diode.
[0145] Meanwhile, the light-emitting element **150** of FIG. **5** has been described as a structure of a general red light-emitting diode in which the first ohmic layer **154** and the p-type layer **151** are sequentially placed on the p-electrode **157** and the second ohmic layer **155**, the n-type layer **153**, the active layer **152**, and the p-type layer **151** are sequentially placed on the n-electrode **158**. Alternatively, the light-emitting element **150** of FIG. **5** may be a general green or blue light-emitting diode in which the first ohmic layer, the p-type layer, the active layer, and n-type layer are sequentially placed on the p-electrode and the second ohmic layer and the n-type layer are sequentially placed on the n-electrode **158**, and in this case, the arrangement and connection of the element may be changed in order to match the first and second electrodes **135** and **136**.
[0146] Here, the light-emitting element **150** may have a concave portion between the p-electrode **157** and the n-electrode **158**. The adhesive layer **140** may be in contact with the protection layer **156** in the concave portion.
[0147] Next, the second flexible substrate FS**2** of FIG. **2** may be provided on the light-emitting element **150** and the stretchable line **134**. The second flexible substrate FS**2** may be attached to the top surfaces of the light-emitting element **150** and the stretchable line **134** through the second adhesive layer.
[0148] The stretchable display device according to the embodiment of the present disclosure can implement a relatively large screen in an unfolded state and a relatively high resolution in a folded state, and this will be described with reference to FIG. **6** and FIG. **7** together with FIG. **1** and FIG. **2**.
[0149] FIG. **1** and FIG. **2** schematically show a stretchable display device according to the embodiment of the present disclosure in the unfolded state, and FIG. **6** and FIG. **7** schematically show the stretchable display device according to the embodiment of the present disclosure in the folded state. Here, FIG. **6** is a schematic plan view of the stretchable display device according to the embodiment of the present disclosure in the folded state, and FIG. **7** is a schematic cross-sectional view corresponding to line III-III′ and line IV-IV′ in FIG. **6**.
[0150] As shown in FIG. **1** and FIG. **2**, in the unfolded state, the first display area DA**1** and the second display area DA**2** may not overlap each other and may be disposed adjacent to each other in the first direction X.
[0151] Light from the light-emitting elements of the first, second, and third rigid portions A**11**, A**12**, and A**13** may be emitted in one direction, for example, in an upward direction toward the second flexible substrate FS**2** and may be output to the outside through the second flexible substrate FS**2**.
[0152] On the other hand, as shown in FIG. **6** and FIG. **7**, in the folded state, the first display area DA**1** and the second display area DA**2** may overlap each other. In this case, the second display area DA**2** may be disposed under the first display area DA**1**. That is, second display area DA**2** may be folded into the rear surface of the first display area DA**1** to thereby overlap each other. The portion of the first flexible substrate FS**1** of the first display area DA**1** and the portion of the first flexible

substrate FS**1** of the second display area DA**2** may be disposed inside to face each other, and the portion of the second flexible substrate FS**2** of the first display area DA**1** and the portion of the second flexible substrate FS**2** of the second display area DA**2** may be exposed to the outside.
[0153] In FIG. **7**, it is shown that the first rigid portion A**11** overlaps the soft portion A**2** of the second display area DA**2** and the second rigid portion A**12** overlaps the soft portion A**2** of the first display area DA**1**. However, the first rigid portion A**11** does not actually overlap the soft portion A**2** of the second display area DA**2**, and the second rigid portion A**12** does not actually overlap the soft portion A**2** of the first display area DA**1**. In addition, it is shown that the emission portion A**1***a* of the third rigid portion A**13** overlaps the soft portion A**2** of the first display area DA**1** and the connection portion A**1***b* overlaps the first rigid portion A**11** of the first display area DA**1**. However, the emission portion A**1***a* of the third rigid portion A**13** does not actually overlap the soft portion A**2** of the first display area DA**1**, and the connection portion A**1***b* does not actually overlap the first rigid portion A**11** of the first display area DA**1** and does overlap the soft portion A**2** of the first display area DA**1**.
[0154] In this case, the first rigid portion A**11** of the first display area DA**1** may be disposed between the adjacent soft portions A**2** of the second display area DA**2**, the second rigid portion A**12** of the second display area DA**2** may be disposed between the adjacent soft portions A**2** of the first display area DA**1**. Accordingly, the second rigid portion A**12** may be disposed between the adjacent first rigid portions A**11** in a third direction or a fourth direction crossing the first and second directions X and Y, and the first rigid portion A**11** between the adjacent second rigid portions A**12** in the third direction or the fourth direction. The first rigid portions A**11** and the second rigid portions A**12** may alternate in the third direction or the fourth direction.
[0155] In addition, the emission portion A**1***a* of the third rigid portion A**13** may be disposed between the adjacent soft portions A**2** of the first display area DA**1** and disposed between the adjacent first rigid portions A**11** in the third direction or the fourth direction.
[0156] Meanwhile, the soft portions A**2** of the first display area DA**1** may overlap the soft portions A**2** of the second display area DA**2**. Specifically, the horizontal soft portion A**21** of the first display area DA**1** may overlap the vertical soft portion A**22** of the second display area DA**2**, and the vertical soft portion A**22** of the first display area DA**1** may overlap the horizontal soft portion A**21** of the second display area DA**2**.
[0157] In the folded state, light from the light-emitting elements of the first rigid portions A**11** may be emitted in the upward direction toward the second flexible substrate FS**2** of the first display area DA**1** and may be output to the outside through the second flexible substrate FS**2**.
[0158] On the other hand, in the folded state, light from the light-emitting elements of the second and third rigid portions A**12** and A**13** may be emitted in a downward direction toward the first flexible substrate FS**1** of the second display area DA**2** and may be output to the outside through the first flexible substrate FS**1** of the second display area DA**2** and the first flexible substrate FS**1** and the second flexible substrate FS**2** of the first display area DA**1**.
[0159] The light-emitting elements of the second and third rigid portions A**12** and A**13** may be double-sided light-emitting diodes, and the light-emitting elements of the first rigid portions A**11** may be one-sided light-emitting diodes. However, embodiments of the present disclosure are not limited thereto.
[0160] In other embodiments, the light-emitting elements of the first rigid portions A**11** may be double-sided light-emitting diodes, and the light-emitting elements of the second and third rigid portions A**12** and A**13** may be one-sided light-emitting diodes. Alternatively, all light-emitting elements of the first, second, and third rigid portions A**11**, A**12**, and A**13** may be double-sided light-emitting diodes.
[0161] In addition, the up and down positions of the first and second display areas DA**1** and DA**2** may be reversed, and the first and second display areas DA**1** and DA**2** may be folded such that the portions of the second flexible substrate FS**2** of the first and second display areas DA**1** and DA**2**

may be disposed inside to face each other and the portions of the first flexible substrate FS**1** of the first and second display areas DA**1** and DA**2** may be exposed to the outside.

[0162] In the folded state, light can be emitted in not only the first rigid portion A**11** but also the second rigid portion A**12** and the third rigid portion A**13** between the adjacent first rigid portions A**11** in the third direction or the fourth direction. Accordingly, the resolution in the folded state may be higher than the resolution in the unfolded state. For example, the resolution in the folded state may be twice the resolution in the unfolded state.

[0163] As such, in the stretchable display device, in the unfolded state, the relatively large screen may be implemented by displaying the image through the first and second display areas DA**1** and DA**2** horizontally adjacent to each other, and in the folded state, the relatively high resolution may be implemented by displaying the image through the first and second display areas DA**1** and DA**2** vertically overlapping each other.

[0164] In this case, the first and second rigid portions A**11** and A**12** of the first and second display areas DA**1** and DA**2** may be spaced apart by ½ pitch in the second direction Y and may have substantially the same configuration, and the emission portion A**1***a* of the third rigid portion A**13** may have substantially the same configuration as the second rigid portion A**12**. Accordingly, the stretchable display device according to the embodiment of the present disclosure can be manufactured using the existing panel process and transfer process.

[0165] In the stretchable display device of the present disclosure, the rigid portions of the first display area and the second display area may be spaced apart by ½ pitch, thereby implementing the relatively large screen in the unfolded state and implementing the relatively high resolution in the folded state.

[0166] In this case, since the rigid portions of the first display area and the second display area have substantially the same configuration, the stretchable display device can be manufactured using the existing panel process and transfer process, so that the process can be optimized or improved, and the production energy can be reduced.

[0167] It will be apparent to those skilled in the art that various modifications and variations can be made in the stretchable display device of the present disclosure without departing from the technical idea or scope of the disclosure. Thus, it is intended that the present disclosure cover the modifications and variations of this disclosure provided they come within the scope of the appended claims and their equivalents.

Claims

1. A stretchable display device, comprising: a display area including a first display area and a second display area adjacent to each other in a first direction; a plurality of rigid portions in the display area and spaced apart from each other in the first direction and a second direction; and a soft portion between adjacent rigid portions, among the plurality of rigid portions, in the first direction and the second direction, wherein the plurality of rigid portions include first, second, and third rigid portions, wherein the first rigid portion is provided in the first display area, and the second rigid portion is provided in the second display area, and wherein the third rigid portion is disposed between the first rigid portion and the second rigid portion and has a larger area than each of the first and second rigid portions.

2. The stretchable display device of claim 1, wherein the first rigid portion and the second rigid portion are disposed on different lines in the first direction.

3. The stretchable display device of claim 2, wherein one rigid portion and one soft portion form a unit, and the first rigid portion and the second rigid portion are spaced apart by ½ pitch of the unit.

4. The stretchable display device of claim 1, wherein the third rigid portion has an L-like shape.

5. The stretchable display device of claim 4, wherein the third rigid portion includes an emission portion and a connection portion, and the emission portion has a same shape and area as the first and second rigid portions.

6. The stretchable display device of claim 5, wherein the emission portion is disposed on a same line as the second rigid portion in the first direction and disposed on a different line from the first rigid portion in the first direction.

7. The stretchable display device of claim 5, wherein the connection portion has a longer length than the emission area in the second direction.

8. The stretchable display device of claim 5, wherein one rigid portion and one soft portion form a unit, and the connection portion has a length greater than ½ pitch of the unit and equal to or smaller than a pitch of the unit.

9. The stretchable display device of claim 5, wherein connection portions of adjacent third rigid portions in the second direction are spaced apart from each other.

10. The stretchable display device of claim 5, wherein: with the stretchable display device not stretched, connection portions of adjacent third rigid portions in the second direction are in contact with each other; and with the stretchable display device stretched, the connection portions of the adjacent third rigid portions in the second direction are spaced apart from each other.

11. The stretchable display device of claim 1, wherein in a folded state in which the first display area overlaps the second display area, the second rigid portion is disposed between adjacent first rigid portions in a third direction or a fourth direction crossing the first direction and the second direction.

12. The stretchable display device of claim 11, wherein a resolution in the folded state in which the first display area overlaps the second display area is greater than a resolution in an unfolded state in which the first display area does not overlap the second display area.

13. The stretchable display device of claim 11, wherein: the soft portion includes a horizontal soft portion extending in the first direction and a vertical soft portion extending in the second direction; and in the folded state in which the first display area overlaps the second display area, the horizontal soft portion of the first display area overlaps the vertical soft portion of the second display area, and the vertical soft portion of the first display area overlaps the horizontal soft portion of the second display area.

14. The stretchable display device of claim 1, further comprising: a pixel provided in each of the plurality of rigid portions and including a plurality of sub-pixels; and a stretchable line provided in the soft portion and connecting adjacent pixels.

15. The stretchable display device of claim 14, wherein: each of a plurality of sub-pixels includes a light-emitting diode, at least one transistor, and at least one capacitor; and the light-emitting diode provided in at least one of the first, second, and third rigid portions is a double-sided light-emitting diode.